

State-Conditioned Gate Arbitration in SAT/CDCL

A Preregistration for MAAT v1.8 Gate Challenge Cycle II

MAAT Structural Selection Series – Paper 75

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2026

Abstract

Paper 74 proposed state-conditioned structural selection after two static SAT constructs failed to receive support in Gate Challenge Cycle I. This paper freezes the corresponding Cycle-II test before any external state-conditioned execution. The decision state X_t , residual-CNF support definitions, normalization, causal updates, rollback, evaluation frequency, calibration sampling, dataset hashes and splits, MOMS fallback, activation budget, no-harm bound, overhead accounting, baselines, ablations, bootstrap procedure, common random numbers, and independent result axes are specified at code level. The primary policy evaluates $H[X_t]$, $B[X_t]$, $S[X_t]$, $V[X_t]$ after every conflict-free unit-propagation fixpoint and activates the frozen Mode-A structural brancher on the highest-budgeted quarter of calibration-normalized states; otherwise it falls back to MOMS. The constants are fixed at $q = 0.25$, relative no-harm bound $\delta = 0.05$, minimum test activation $q_{\text{test},\min} = 0.125$, distinctness tolerance $\varepsilon = 0.01$, reconstruction threshold $\rho_\star = 0.90$, 5000 conflicts, 30 seconds per policy-instance, and 10,000 instance-level bootstrap resamples. Uncertainty is secondary and cannot alter the primary gate. The frozen manifest contains metadata and SHA256 hashes for 98 public SATLIB instances, split into 19 calibration and 79 test instances; raw CNFs are excluded. No external state-conditioned benchmark is executed here. Paper 76 is prohibited until this exact preregistration has a public Zenodo DOI.

1 Status, Scope, and Release Barrier

This is a preregistration, not an execution paper. It converts the Paper-74 framework hypothesis [10] into one independently executable SAT/CDCL protocol. It does not change the Paper-68 record [6], reinterpret the Paper-69 negative result [7], or repair the degenerate Paper-72 channel after its Paper-73 execution [8, 9]. The chronology is explicit: the Series-II archive DOI <https://doi.org/10.5281/zenodo.21062386> covers Papers 69–71; Paper 72 was separately pre-registered before execution at <https://doi.org/10.5281/zenodo.21063444>, and Paper 73 was archived at <https://doi.org/10.5281/zenodo.21123988>.

Before the public Paper-75 Zenodo release, only synthetic or explicitly excluded development fixtures may exercise the implementation. No external state-conditioned calibration, threshold estimation, policy comparison, or utility measurement is allowed. Paper 76 may begin only after the exact protocol, schema, manifest, and tests described here have a public DOI.

The hypothesis is deliberately narrow:

Evaluating the existing structural support language on the current CDCL decision state, and arbitrating against a strong classical fallback, can add decision utility beyond static MAAT scoring and gating.

The null is that it cannot do so under the frozen rules.

2 Exact Decision State

Let t index branch-decision boundaries, not propagations or wall-clock ticks. The state is

$$X_t = (\mathcal{C}_0, \mathcal{C}_t^{\text{learn}}, A_t, L_t, T_t, Q_t, \mathbf{a}_t^{\text{VSIDS}}, \phi_t), \quad (1)$$

where $\mathcal{C}_0$ is the original clause set, $\mathcal{C}_t^{\text{learn}}$ the persistent learned clauses, A_t the partial assignment, L_t decision levels, T_t the trail, Q_t reason clauses, $\mathbf{a}_t^{\text{VSIDS}}$ activity, and ϕ_t saved phase.

The residual formula F_t is constructed by dropping every satisfied clause from $\mathcal{C}_0 \cup \mathcal{C}_t^{\text{learn}}$ and removing every falsified literal from the remaining clauses. Evaluation occurs exactly once after unit propagation reaches a conflict-free fixpoint and immediately before the next branch decision, including the root decision. Conflicting states and fully satisfied states are not gate states. The primary transparent backend uses no learned-clause deletion. These conventions fix what information can enter a support and prevent look-ahead leakage.

3 Code-Level State Supports

Let U_t be the unassigned variables appearing in at least one nonempty clause of F_t . For $v \in U_t$, let p_v and n_v count positive and negative residual occurrences, and let $d_v = p_v + n_v$. Set $\varepsilon_0 = 10^{-9}$.

3.1 Consistency and Balance

The literal-polarity inconsistency is

$$I_t = \frac{1}{|U_t|} \sum_{v \in U_t} \frac{|p_v - n_v|}{p_v + n_v + \varepsilon_0}, \quad H[X_t] = \frac{1}{1 + I_t}. \quad (2)$$

The occurrence-pressure coefficient of variation and balance support are

$$c_d = \frac{\text{std}_{v \in U_t}(d_v)}{\text{mean}_{v \in U_t}(d_v) + \varepsilon_0}, \quad B[X_t] = \frac{1}{1 + c_d}. \quad (3)$$

3.2 Controlled Activity

For each active variable define residual short-clause pressure

$$s_v = \sum_{\substack{C \in F_t: v \in C \\ |C| \leq 3}} \frac{1}{|C|}. \quad (4)$$

The activity support measures how broadly this pressure is distributed:

$$S[X_t] = \begin{cases} \frac{\text{mean}_{v \in U_t}(s_v)}{\max_{v \in U_t} s_v + \varepsilon_0}, & \max_v s_v > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Thus a single short-clause hotspot does not automatically count as globally supported activity.

3.3 Connectivity

Construct the residual variable co-occurrence graph on U_t , joining two variables when they share an active clause. Let f_{LCC} be its largest connected-component fraction and $\bar{c}$ its mean local clustering. Then

$$V[X_t] = \text{clip}_{[0,1]} (0.65f_{\text{LCC}} + 0.35\bar{c}). \quad (6)$$

All four supports are clipped to $[10^{-9}, 1]$ before closure. Robustness is not a fifth primitive support:

$$R_{\text{resp}}[X_t] = (HBV)^{1/3}, \quad (7)$$

$$R_{\text{struct}}[X_t] = (HBSV)^{1/4}, \quad (8)$$

$$R_{\text{rob}}[X_t] = \min(R_{\text{resp}}[X_t], R_{\text{struct}}[X_t]). \quad (9)$$

4 Normalization and Frozen Gate

SAT polarity is $p_D = +1$. Calibration-state weighted medians and median absolute deviations define

$$z_R[X_t] = \frac{R_{\text{rob}}[X_t] - \text{med}_{\text{cal}}(R_{\text{rob}})}{\text{MAD}_{\text{cal}}(R_{\text{rob}}) + 10^{-9}}, \quad (10)$$

$$\Delta R_t = R_{\text{rob}}[X_t] - R_{\text{rob}}[X_{t-1}], \quad (11)$$

$$z_t[X_t] = \frac{\Delta R_t - \text{med}_{\text{cal}}(\Delta R)}{\text{MAD}_{\text{cal}}(\Delta R) + 10^{-9}}. \quad (12)$$

At the root, $\Delta R_0 = 0$. No canonical SAT spatial derivative was frozen, so the Paper-68 missing-response rule is applied: $z_x = 0$. The unchanged response-gate coefficients give

$$g_t = z_R + 0.50z_t + 0.25z_x, \quad G_{\text{gate}}[X_t] = \frac{1}{1 + e^{-g_t}}. \quad (13)$$

The activation budget is fixed at

$$q = 0.25. \quad (14)$$

Using family-, instance-, and state-balanced calibration weights, θ_q is the weighted 0.75-quantile of G_{gate} , with lower interpolation. Let $W_>$ and $W_=>$ be calibration weight above and exactly at θ_q . Freeze

$$f_{\text{tie}} = \text{clip}_{[0,1]} \frac{q - W_>}{W_=>}, \quad (15)$$

with $f_{\text{tie}} = 0$ when $W_=>=0$. The primary gate opens for $G > \theta_q$, closes for $G < \theta_q$, and at equality opens only when the state-bound `gate_tie` hash is below f_{tie} . Both θ_q and f_{tie} are frozen from calibration. No test-state refit is allowed. The Paper-68 hard threshold 0.5 remains an ablation only.

5 Arbitrated Policy

The fallback is MOMS [2]. On activation, the policy uses the frozen Paper-63 Mode-A brancher [5] on the current frontier:

$$\pi_{75}(t) = \begin{cases} \pi_{\text{ModeA}}(X_t), & \text{the frozen budget gate activates,} \\ \pi_{\text{MOMS}}(X_t), & \text{otherwise.} \end{cases} \quad (16)$$

Candidate score is

$$A_t(v) = P_t(v) - 0.38 h_{\text{MAAT},t}(v), \quad (17)$$

sampled with softmax inverse temperature $\beta = 8$. Every directly compared stochastic primary policy uses the same state-bound random number

$$u_{i,t} = \text{HashToUnit}(\text{paper75}|75075|\text{primary}|\text{dataset_id}|t|\text{state_fingerprint}). \quad (18)$$

Policy identity is excluded. Null and debug policies use separate declared namespaces. The progress term is

$$P_t(v) = 0.52 \text{VSIDS}_v^{\text{norm}} + 0.26 \text{JW}_v^{\text{norm}} + 0.14 u_v^{\text{norm}} + 0.08 d_v^{\text{norm}}, \quad (19)$$

where u_v is binary-clause unit-gain pressure. Candidate-local h_{MAAT} is the frontier min-max normalization of $-\sum_{a \in \{H,B,S,V,R_{\text{rob}}\}} \log(10^{-9} + a_v)$, following the Paper-63 implementation. Literal polarity is chosen by the larger combined occurrence, Jeroslow-Wang [3], and unit-gain score; saved phase is reused with probability 0.25, as in Paper 69.

6 Causal Updates, Rollback, and Frequency

The implementation may access only current and past solver state. It may not use trial assignments, rollouts, labels, solved status, or test-set aggregates.

- **Enqueue:** update assignment, clause status, residual literal counters, graph-edge multiplicities, and short-clause counters; log inverse deltas on the trail.
- **Learn:** insert a learned clause after conflict analysis. It persists across backtracking.
- **Rollback:** undo assignment-induced deltas in reverse trail order above the target level; do not remove learned clauses.
- **Audit:** synthetic tests check every event. Paper 76 must compare incremental supports with full recomputation every 100th decision and abort on absolute discrepancy above 10^{-12} .

Supports and the gate are evaluated once per branch-decision boundary, after propagation and before candidate scoring. No evaluation is triggered by an individual propagation or during conflict analysis.

7 Calibration-State Sampling and Weighting

All calibration states are generated exclusively by frozen MOMS-only traces under the Paper-75 conflict/runtime budgets and seed rules. No state produced by v1.8 arbitration, another evaluated policy, or a test instance enters normalization, thresholding, distinctness, reconstruction, or overhead conversion.

Each calibration instance retains at most 128 evaluated states. If more are encountered, retain the 128 smallest deterministic SHA256 priorities of

$$\text{paper75}|\text{dataset_id}|\text{decision_index}|\text{state_fingerprint}.$$

The state fingerprint hashes decision level, trail, and sorted residual clauses. If fewer than 128 states exist, retain all.

Each family has total weight $1/n_{\text{families}}$, each instance has equal weight within its family, and each retained state has equal weight within its instance. Hence a long or easy trajectory cannot dominate calibration merely by producing more states. Calibration runs determine normalization, θ_q , and the runtime conversion constant; they do not contribute to test utility.

8 Dataset, Hashes, and Splits

The frozen metadata manifest contains 98 instances from six public SATLIB families [1]: AIM (20), Dubois (13), pigeonhole (5), flat graph colouring (20), satisfiable random 3-SAT UF20 (20), and unsatisfiable random 3-SAT UUF50 (20). Within each family, rows are sorted by

$$\text{SHA256}(\text{paper75}|75075|\text{dataset_id}|\text{instance_sha256}). \quad (20)$$

The first $\max(1, \lfloor 0.20n_f \rfloor)$ rows are calibration; the remainder are test. This yields 19 calibration and 79 test instances.

The frozen manifest SHA256 is

$$941fad18c5ea4839ee3aac4b5d7f1e3d5dd0cf8ed8c8f37f3ec9a4a9a74df3ad.$$

Paper 76 must verify existence, exact file hash, DIMACS parseability, and the absence of unknown CNFs before any solver run. Failure produces `dataset_gatekeeper_fail`. Raw CNFs and archives are not redistributed.

9 Budgets and Overhead Accounting

The per-policy, per-instance limits are 5000 conflicts and 30 seconds. Search cost is

$$C_{\text{search}} = N_{\text{conflict}} + 0.30N_{\text{decision}} + 0.004N_{\text{propagation}} + 5000 \mathbf{1}[\text{timeout}]. \quad (21)$$

Structural update, support evaluation, gate calculation, and arbitration are timed separately with a monotonic nanosecond clock. Let

$$\kappa_{\text{cal}} = \frac{\sum_{i \in \text{cal}} C_{\text{search}}^{\text{MOMS}}(i)}{\sum_{i \in \text{cal}} T_{\text{wall}}^{\text{MOMS}}(i)}. \quad (22)$$

Primary total cost includes structural overhead:

$$C_{\text{total}} = C_{\text{search}} + \kappa_{\text{cal}} T_{\text{structural}}. \quad (23)$$

Wall runtime, which also includes all overhead, is reported separately.

For the primary fallback comparison, relative regret is

$$r_i = \frac{C_{75}(i) - C_{\text{MOMS}}(i)}{\max(C_{\text{MOMS}}(i), 10^{-9})}. \quad (24)$$

The no-harm bound is frozen at $\delta = 0.05$, but uncertainty is not called harm. Let $[\ell_r, u_r]$ be the paired 95% bootstrap interval of mean relative regret. The independent safety status is

$$\begin{cases} \text{no_harm_certified}, & u_r \leq 0.05, \\ \text{harmful}, & \ell_r > 0.05, \\ \text{no_harm_not_established}, & \ell_r \leq 0.05 < u_r. \end{cases} \quad (25)$$

10 Distinctness and Reconstruction Preconditions

Distinctness tolerance is $\varepsilon = 0.01$. On retained calibration states, the construction is declared `not_testable_as_constructed` if any pair among H, B, S, V has

$$1 - |\rho_{\text{Spearman}}| < 0.01 \quad (26)$$

or if any support has weighted standard deviation below 10^{-8} .

The classical-reconstruction check predicts each of $H, B, S, V, R_{\text{rob}}, G_{\text{gate}}$ from three candidate vectors. For variable v , these are MOMS $m_v = p_v + n_v + 2p_v n_v$ on shortest residual clauses, Jeroslow–Wang $j_v = j_v^+ + j_v^-$ with clause weight $2^{-|C|}$, and current raw VSIDS activity a_v . Each vector contributes exactly six scalars:

$$\max, \quad \text{mean}, \quad \text{std}, \quad \frac{\text{std}}{|\text{mean}| + 10^{-9}}, \quad \frac{s_{(1)} - s_{(2)}}{|s_{(1)}| + 10^{-9}}, \quad \frac{-\sum_v p_v \log p_v}{\log n}. \quad (27)$$

The top-two gap is zero for fewer than two candidates. Normalized entropy is zero for $n \leq 1$ or zero nonnegative score mass. Thus the OLS design has 18 predictors plus an intercept. Five-fold cross-validation is grouped by calibration instance and deterministically hashed with seed 75075. The primary gate coordinates are R_{rob} and G_{gate} ; component supports are reported but do not independently trigger this outcome. If the family- and instance-weighted out-of-fold R^2 of either primary coordinate reaches

$$\rho_\star = 0.90, \quad (28)$$

the leading interpretation is `convergent_rediscovery`, not novel structural information.

11 Baselines, Nulls, and Ablations

Required classical baselines are MOMS, Jeroslow–Wang, and VSIDS [2, 3, 4]. Required MAAT baselines are Paper-69 score-with- R , the static v1.7 gate, and the primary state-conditioned arbitration. Nulls are a gate-signal shuffle within family and split, and random activation matched to $q = 0.25$. For the shuffle, record G -sequences on the required MOMS baseline trace; sort instances by $\text{SHA256}(75075|\text{dataset_id})$, cyclically assign each instance the next donor in its family and split, and at decision t use $G_{\text{donor}}[t \bmod n_{\text{donor}}]$ with the same θ_q . The random null activates when the 256-bit integer represented by $\text{SHA256}(75075|\text{dataset_id}|t|\text{random_null})$ is below $0.25(2^{256} - 1)$. Neither null uses outcomes or utilities.

Frozen secondary ablations are: default-only MOMS; always-active state Mode A; value-only z_R ; response-only $0.50z_t$; full $z_R + 0.50z_t$; the Paper-68 hard threshold 0.5; and leave-one-support closure variants HBV, HSV, BSV, and HBS. Ablations cannot replace the primary policy or alter its classification.

12 Bootstrap and Secondary Uncertainty

For policy A against comparator B , define

$$\Delta U_i(A, B) = C_B(i) - C_A(i), \quad (29)$$

so positive values favour A . Paper 76 must perform 10,000 paired instance-level bootstrap resamples with seed 75075: resample instances within each family, aggregate each family equally, then average families. Report mean delta, 95% interval, wins/losses/ties, family deltas, activation rate, overhead fraction, and timeouts.

Uncertainty in θ_q , support bands, and frozen-threshold gate margins is reported as a secondary analysis. It may not change q , θ_q , f_{tie} , the gate equation, or any status axis. No uncertainty-aware retuning is permitted in Paper 76.

13 Complete Status System

One exclusive label is insufficient: reconstruction, activation, safety, and utility can coexist. Paper 76 therefore reports every status axis. A failed dataset gatekeeper prevents execution; failed distinctness prevents primary utility execution. Otherwise construct status never suppresses safety or utility.

Axis	Status	Frozen rule
Execution	<code>dataset_gatekeeper_fail</code>	Any manifest, hash, parseability, or unknown-file check fails; no execution.
Execution	<code>executed</code>	Gatekeeper passes and the primary run completes.
Construct	<code>distinct</code>	Distinctness passes and both primary gate coordinates have reconstruction $R^2 < 0.90$.
Construct	<code>not_testable_as_constructed</code>	Distinctness fails; primary utility execution stops.
Construct	<code>convergent_rediscovery</code>	R_{rob} or G has weighted out-of-fold $R^2 \geq 0.90$; utility remains independently reportable.
Activation	<code>adequate_activation</code>	Family-instance-weighted test decision activation rate is at least $q_{\text{test,min}} = 0.125$.
Activation	<code>activation_collapse</code>	Test activation is below 0.125; routing and strong support are forbidden.
Safety	<code>no_harm_certified</code>	Relative-regret CI upper bound is at most 0.05.
Safety	<code>no_harm_not_established</code>	CI spans the 0.05 bound.
Safety	<code>harmful</code>	Relative-regret CI lower bound exceeds 0.05.
Utility	<code>negative_result</code>	CI upper bound is non-positive against static v1.7 or score-with- R .
Utility	<code>inconclusive</code>	Minimum support is not met and the negative criterion does not apply.
Utility	<code>minimum_support</code>	CI lower bounds are positive against static v1.7, score-with- R , and both matched-rate nulls.
Utility	<code>routing_support</code>	Minimum support, adequate activation, no-harm certification, and positive CI lower bound versus MOMS among test instances whose activation rate is at least 0.125.
Utility	<code>strong_sat_support</code>	Minimum support, adequate activation, no-harm certification, and positive overall CI lower bounds versus MOMS and Jeroslow–Wang.

No other status label is permitted. In particular, a secondary ablation cannot promote primary utility, and `convergent_rediscovery` cannot hide a harmful or negative utility result.

14 Machine-Readable Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Planned protocol folder:

`experiments/maat_v18_state_conditioned_prereg/`

The release contains the TeX/PDF, frozen JSON, JSON Schema, metadata/hash manifest, dependency-free validator, pure state-semantics module, README, and synthetic unit tests. The commands are

```
python3 paper75_preregistration.py
python3 validate_preregistration.py
python3 -m unittest discover -s tests -v
```

The validator must end with `VALIDATION PASS`. No solver runner or external result file belongs to Paper 75.

15 Conclusion

Paper 75 freezes the first test of the unresolved degree of freedom identified after Gate Challenge Cycle I: whether structural supports become useful when evaluated on the same evolving residual state as their strongest classical competitors. The protocol gives that hypothesis a fair route to support, but also several ways to fail: degeneracy, classical reconstruction, harm, negative utility, or inconclusive intervals. All are informative outcomes. The next scientific act is not another definition. It is public archiving of this protocol, followed by one unchanged Paper-76 execution.

References

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